

● MEDIUM

● ALGEBRA

Systems of two linear equations in two variables

01

10 questions · ~15 min

Q1

ALGEBRA

$$x + 2y = 6$$

$$x - 2y = 4$$

The solution to the given system of equations is x, y . What is the value of x ?

- A. 2.5
- B. 5
- C. 6
- D. 10

Q2

ALGEBRA

Two customers purchased the same kind of bread and eggs at a store. The first customer paid 12.45 dollars for 1 loaf of bread and 2 dozen eggs. The second customer paid 19.42 dollars for 4 loaves of bread and 1 dozen eggs. What is the cost, in dollars, of 1 dozen eggs?

- A. 3.77
- B. 3.88
- C. 4.15
- D. 4.34

Q3

ALGEBRA

At how many points do the graphs of the equations $y = x + 20$ and $y = 8x$ intersect in the xy -plane?

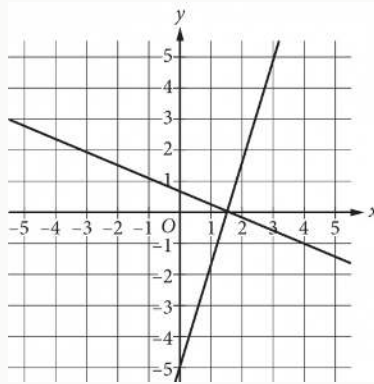
- A. 0
- B. 1
- C. 2
- D. 8

Q4

ALGEBRA

In the xy -plane, the graph of $y = x + 3$ intersects the graph of $y = 2x - 6$ at the point (a, b) . What is the value of a ?

- A. 3
- B. 6
- C. 9
- D. 12



Which of the following systems of equations has the same solution as the system of equations graphed above?

A.
$$\begin{aligned} y &= 0 \\ x &= \frac{3}{2} \end{aligned}$$

B.
$$\begin{aligned} y &= \frac{3}{2} \\ x &= 0 \end{aligned}$$

C.
$$\begin{aligned} y &= 0 \\ x &= 1 \end{aligned}$$

D.
$$\begin{aligned} y &= 1 \\ x &= 0 \end{aligned}$$

$$6x + 7y = 28$$

$$2x + 2y = 10$$

The solution to the given system of equations is x, y . What is the value of y ?

A. -2

B. 7

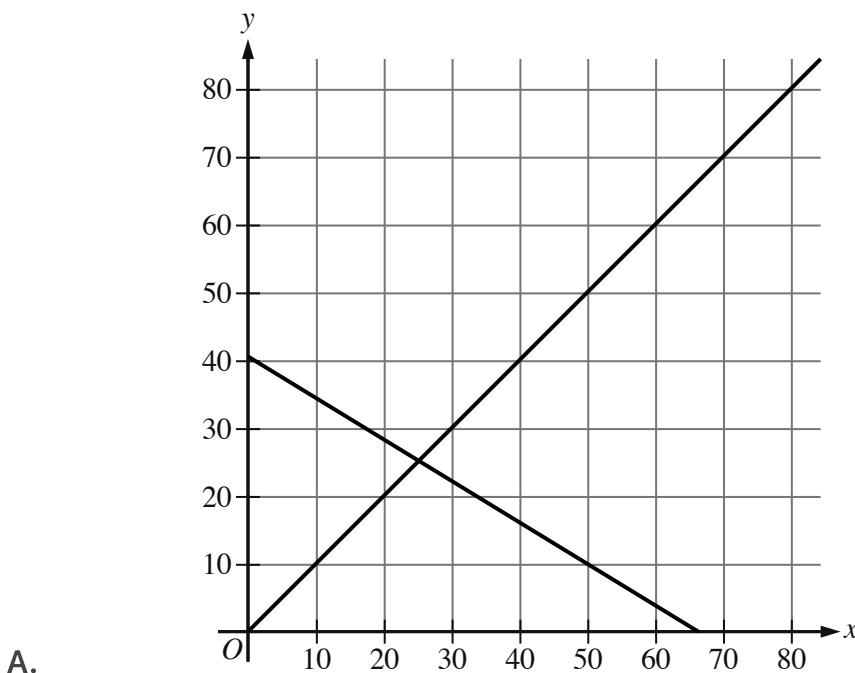
C. 14

D. 18

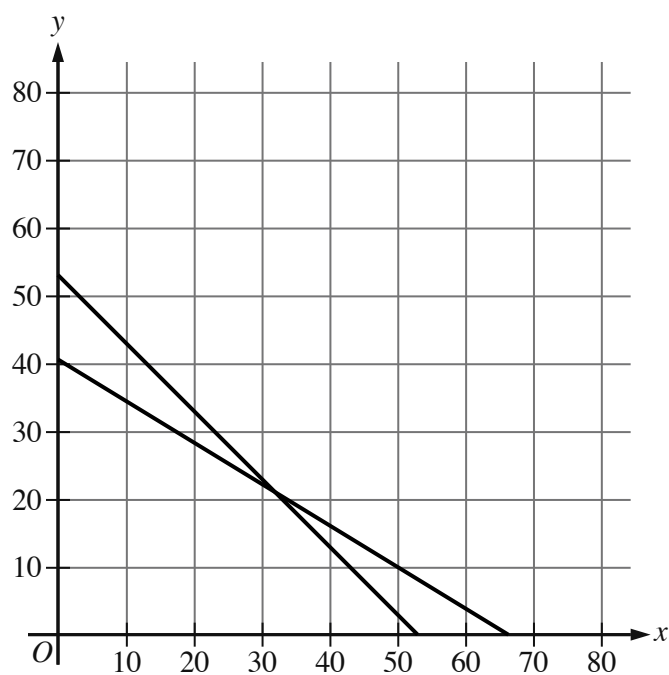
$$x + y = 53$$

$$11x + 18y = 730$$

The given equations represent the possible numbers of beach chairs, x , and umbrellas, y , rented at a park last month and the total spent, in dollars, to rent those beach chairs and umbrellas. Which of the following graphs represents this situation?



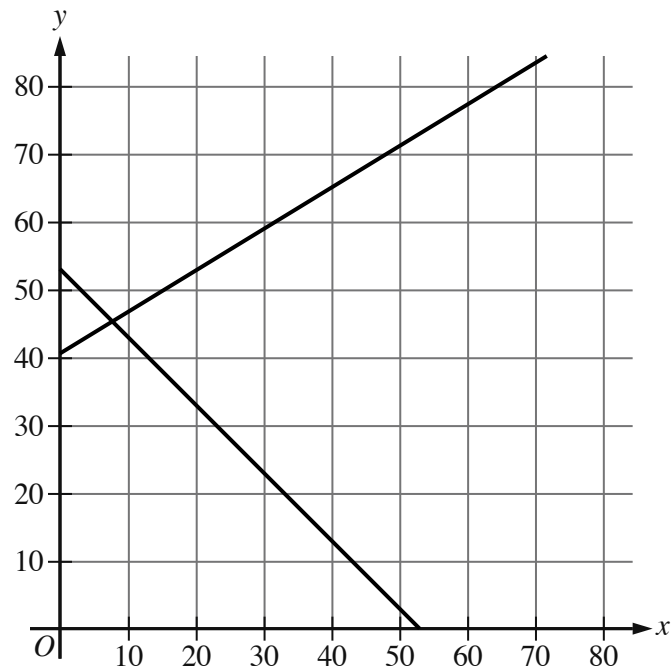
- For the first line in the system:
 - The line slants gradually down from left to right.
 - The line passes through the following approximate points:
 - (0 comma 41)
 - (66 comma 0)
- For the second line in the system:
 - The line slants gradually up from left to right.
 - The line passes through the following points:
 - (0 comma 0)
 - (50 comma 50)
- The 2 lines intersect at 1 point.



B.

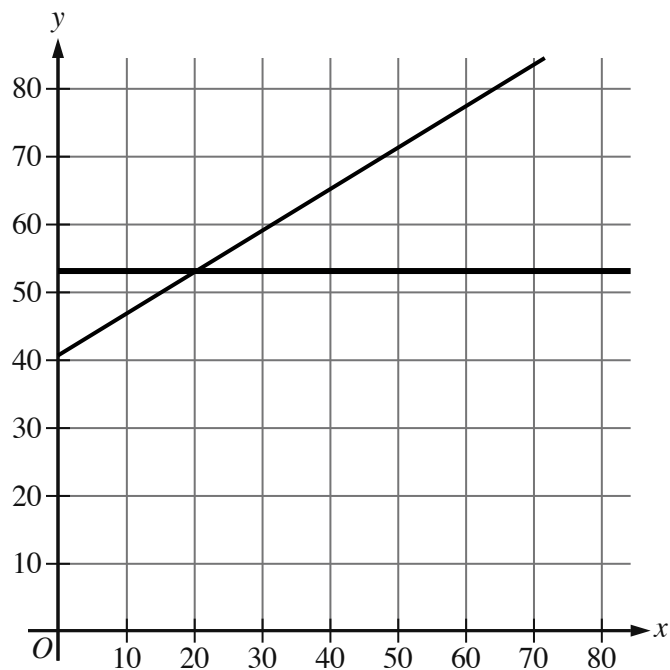
- For the first line in the system:
 - The line slants gradually down from left to right.
 - The line passes through the following points:
 - (0 comma 53)
 - (53 comma 0)
- For the second line in the system:
 - The line slants gradually down from left to right.
 - The line passes through the following approximate points:
 - (0 comma 41)
 - (66 comma 0)
- The 2 lines intersect at 1 point.

C.



- For the first line in the system:
 - The line slants gradually down from left to right.
 - The line passes through the following points:
 - (0 comma 53)
 - (53 comma 0)
- For the second line in the system:
 - The line slants gradually up from left to right.
 - The line passes through the following approximate points:
 - (0 comma 41)
 - (30 comma 59)
- The 2 lines intersect at 1 point.

D.



- For the first line in the system:
 - The line is horizontal.
 - The line passes through the following points:
 - (0 comma 53)
 - (30 comma 53)
- For the second line in the system:
 - The line slants gradually up from left to right.
 - The line passes through the following approximate points:
 - (0 comma 41)
 - (30 comma 59)
- The 2 lines intersect at 1 point.

Q8

GRID-IN

ALGEBRA

$$y = -2x$$

$$3x + y = 40$$

The solution to the given system of equations is x, y . What is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

GRID-IN

ALGEBRA

$$3x + 6 = 4y$$

$$3x + 4 = 2y$$

The solution to the given system of equations is x, y . What is the value of y ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q10

GRID-IN

ALGEBRA

$$y = \frac{1}{3}x - 14$$

$$y = -x + 18$$

The solution to the given system of equations is x, y . What is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.